

Example 1: from a pack of cards, a card is drawn. What is the probability of drawing a red card or a king?

Solution: there are 52 cards. One card can be drawn in C_1^{52} ways. So the total number of event points are C_1^{52} . There are two red suits. Hence, total 26 red cards are there. Again there is a king in each suit. So, total 4 kings are there.

Let, A denote the event of drawing of red card
and B denote the event of drawing a king.

Now, $P(A \cup B) = P(A) + P(B) - P(AB)$

$$P(A) = \frac{C_1^{26}}{C_1^{52}} = \frac{26}{52} = \frac{1}{2}$$

$$P(B) = \frac{C_1^4}{C_1^{52}} = \frac{4}{52} = \frac{1}{13}$$

$$P(AB) = \frac{C_1^2}{C_1^{52}} = \frac{2}{52} = \frac{1}{26}$$

$$\text{Therefore, } P(A \cup B) = \frac{1}{2} + \frac{1}{13} - \frac{1}{26} = \frac{7}{13}$$

Hence, the probability of drawing a red card or a king is $\frac{7}{13}$.

Example 2: A binary communication channel carries data as one of two types of signal denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1 and a transmitted 1 is sometimes received as 0. For a given channel, assume a probability of 0.94 that a transmitted 0 is correctly received as a 0 and a probability of 0.91 that a transmitted 1 is received as 1. Further assume, a probability of 0.45 of transmitting a 0. If a signal is sent, determine-

1. Probability that a 1 is received
2. Probability that a 0 is received
3. Probability that a 1 was transmitted, given that a 1 was received
4. Probability that a 0 was transmitted, given that a 0 was received
5. Probability of an error

Solution: let T_0 denote the event that a 0 is transmitted

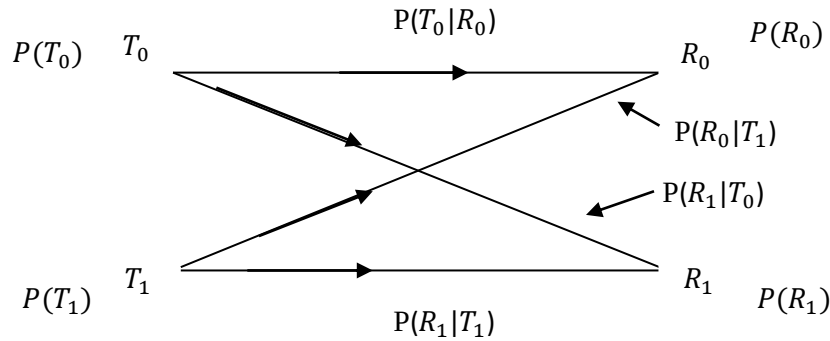
R_0 denotes the event that a 0 is received

$T_1 = \bar{T}_0$ denotes the event that 1 is transmitted

$R_1 = \bar{R}_0$ denotes the event that 1 is received

Therefore, $R_0, R_1, T_1 | R_1, T_0 | R_0$ are the events of interests.

An error occurs when 1 is transmitted but 0 is received or when 1 is received but 0 is transmitted. The event of occurrence of an error can be denoted as $(T_1 \cap R_0) \cup (T_0 \cap R_1)$



Given that, $P(R_0|T_0) = 0.94$ and $P(R_1|T_1) = 0.91$

Also, $P(T_0) = 0.45$

Therefore,

$$P(R_1|T_0) = P(\bar{R}_0|T_0) = 1 - P(R_0|T_0) = 1 - 0.94 = 0.06$$

$$P(R_0|T_1) = P(\bar{R}_1|T_1) = 1 - P(R_1|T_1) = 1 - 0.91 = 0.09$$

Probability of transmitting 1 is,

$$P(T_1) = P(\bar{T}_0) = 1 - P(T_0) = 0.55$$

Probability of **receiving 0** is,

$$P(R_0) = P(R_0|T_0)P(T_0) + P(R_0|T_1)P(T_1) = 0.94 \times 0.45 + 0.09 \times 0.55 = 0.4725$$

Probability of **receiving 1** is,

$$P(R_1) = P(\bar{R}_0) = 1 - P(R_0) = 1 - 0.4725 = 0.5275$$

Therefore, using Bayes theorem, the probability that a **1 was transmitted, given that a 1 was received**, can be calculated as follows:

$$P(T_1|R_1) = \frac{P(R_1|T_1)P(T_1)}{P(R_1)} = \frac{0.91 \times 0.55}{0.5275} = 0.9488$$

Similarly, Probability that a **0 was transmitted, given that a 0 was received**, can be calculated as follows:

$$P(T_0|R_0) = \frac{P(R_0|T_0)P(T_0)}{P(R_0)} = \frac{0.94 \times 0.45}{0.4725} = 0.8952$$

Now,

$$P(T_1|R_0) = P(\bar{T}_0|R_0) = 1 - P(T_0|R_0) = 1 - 0.8952 = 0.1048$$

$$P(T_0|R_1) = P(\bar{T}_1|R_1) = 1 - P(T_1|R_1) = 1 - 0.9488 = 0.0512$$

Finally, the probability of obtaining an error is,

$$\begin{aligned} P((T_1 \cap R_0) \cup (T_0 \cap R_1)) &= P(T_1 \cap R_0) + P(T_0 \cap R_1) = P(T_1|R_0)P(R_0) + P(T_0|R_1)P(R_1) \\ &= 0.0765 \end{aligned}$$

Example 3: a certain item is manufactured by 3 factories say 1,2 and 3. It is known that 1 turns out twice as many items as 2, and that 2 and 3 turns the same number of items (during specified product period). It is also known that 2% of the items produced by 1 and by 2 are defective, while 4% of these manufactured by 3 are defective. All these items produced are put into one stockpile and then one item is chosen at random. What is the probability that the chosen item is defective?

Solution: let us introduce the following events:

$$A = \{\text{The item is defective}\}$$

$$B_1 = \{\text{The item came from 1}\}$$

$$B_2 = \{\text{The item came from 2}\}$$

$$B_3 = \{\text{The item came from 3}\}$$

We require $P(A)$ and we can write:

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3)$$

$$\text{Here, } P(B_1) = \frac{1}{2} \text{ and } P(B_2) = P(B_3) = \frac{1}{4}$$

$$\text{Also, } P(A|B_1) = 0.02 = P(A|B_2) \text{ while, } P(A|B_3) = 0.04$$

$$\text{Hence, } P(A) = 0.02 \times \frac{1}{2} + 0.02 \times \frac{1}{4} + 0.04 \times \frac{1}{4} = 0.025$$

Example 4: An article manufactured by a company consists of two-part A and B. In the process of manufacturing of part A, 9 out of 100 are likely to be defective. Similarly, 5 out of 100 are likely to be defective in the manufacture of part B. Calculate the probability that the assembled article will be defective.

Solution: Let us consider two event A & B as

A= event that part A is not defective and

B= event that part B is not defective

Probability that the assembled article will not be defective i.e., we have to calculate the compound probability $P(AB)$.

Part A I not defective as well as Part B is not defective.

$P(A)$ = probability that part A is not defective

$$= (1 - \text{probability that part A is defective}) = 1 - 9/100 = 0.91$$

$P(B)$ = probability that part B is not defective

$$= 1 - \text{probability that part B is defective} = 1 - 5/100 = 0.95$$

$$P(AB) = P(A) \times P(B) = 0.91 \times 0.95 = .8645$$

Example 5: The odds against Manager X settling the wage dispute with the workers are 8:6 and odds in favor of Manager Y settling the same dispute are 14:16

(i) What is the chance that neither settles the dispute, if they both try independently of each other?

(ii) What is the probability that the dispute will be settled?

Solution: Let A be event that the manager X will settle the dispute and B be the event that the manager Y will settle the dispute.

$$P(A^c) = 8/14 = 4/7, P(A) = (1 - 4/7) = 3/7$$

$$P(B^c) = 16/16 + 14 = 16/30 = 8/15$$

$$P(B) = 1 - 8/15 = 7/15$$

The probability that neither settles the dispute is given by

$$P(A^c \cap B^c) = P(A^c) \cdot P(B^c) = \frac{4}{7} \cdot \frac{8}{15} = \frac{32}{105}$$

(ii) The dispute will be settled if at least one of the managers X and Y settles the dispute i.e.,

$$P(A \cup B) = 1 - P(A^c \cap B^c) = 1 - \frac{32}{105} = \frac{73}{105}$$

Example 6: In answering a question on a multiple choice test a student either knows the answer or he guesses. Let p be the probability that he knows the answer and (1-p) be the probability that he guesses. Assume that a student who guesses at the answer will be correct with probability 1/5, where 5 is the number of multiple-choice alternatives. What is the condition at the probability that a student know the answer to a question given that he answered correctly?

Solutions: Let E_1 be the event that the student knew the correct answer.

E_2 be the event that the student guesses the right answer

And A be the event that the student gets the right answer

Hence, $P(E_1) = p$, $P(E_2) = (1-p)$, and $P(A/E_2) = 1/5$.

$P(A/E_1) = \text{prob}(\text{student gets the right answer given that he knew the right answer}) = 1$ We have to calculate $P(E_1/A)$

$$P(E_1/A) = \frac{(P(E_1).P(A/E_1))}{((P(E_1).P(A/E_1)) + (P(E_2).P(A/E_2)))}$$

$$= \frac{p.1}{p.1 + (1-p).\frac{1}{5}} = \frac{5p}{4p+1}$$

Example 7: In a bolt factory, the machines y_1 , y_2 , and y_3 manufacture respectively 25, 35, and 40 percent of the total product. Of their output 5, 4, and 2 percent respectively are defective bolts. One bolt is drawn from the product and is found to be defective. What is the probability that it was manufactured by machine y_3 ?

Solution: A be the event that selected bolt is defective, B_1 be the event that the selected bolt come from machine y_1 , B_2 be the event that the selected bolt come from machine y_2 , B_3 be the event that the selected bolt come from machine y_3 .

Find out $P(B_3/A)$

$$P(B_1) = 25/100 = 1/4, P(B_2) = 35\% = 35/100 = 7/20, P(B_3) = 40/100 = 2/5.$$

$$P(A/B_1) = 5\% = 5/100 = .05, P(A/B_2) = 4\% = 4/100 = .04, P(A/B_3) = 2\% = .02$$

$$P(B_3/A) = \frac{P(B_3).P(A/B_3)}{P(B_1).P(A/B_1) + P(B_2).P(A/B_2) + P(B_3).P(A/B_3)}$$

$$= \frac{.4 \times .02}{.25 \times .05 + .35 \times .04 + .4 \times .02} = \frac{.008}{.0125 + .0140 + .008} = \frac{16}{69}$$

Exercise 1: in the following two figures we assume that the probability of each relay being closed is p and that each relay is open or closed independently of other relay. In each case find the probability that current flows from L to R.

